

# Year 8 Science

## 2022

### Topic Test: Cells and Microscopes (Weighting 25%)



Name: MARKING KEY

Marks: 55



#### Materials Required:

- Blue/black ballpoint pen
- Pencil
- Ruler
- Eraser
- Calculator

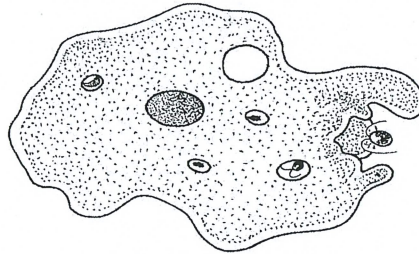
| Section 1<br>(10) | Section 2<br>(45) | Total<br>(55) | Percentage |
|-------------------|-------------------|---------------|------------|
|                   |                   |               |            |

## Section One: Multiple Choice

[10 Marks]

Circle the letter of the most correct answer for each of the statements or questions below.

1. Which of the following statements about cell theory is **false**?
  - a. Cells are the basic units of life
  - b. A virus is a type of cell
  - c. New cells are created from existing cells
  - d. All living things are composed of cells
2. A student looking at a droplet of pond water through her microscope, saw the following object moving around on her slide.



The object is most probably

- a. a cell from a multicellular animal
  - b. a cell from a multicellular plant
  - c. a unicellular plant
  - d. a unicellular animal
3. Which of the following cell components may be found in both plant and animal cells?
  - a. Chloroplasts
  - b. Mitochondria
  - c. Several small vacuoles
  - d. Cell wall
4. Which of the cell organelles below serves as a transport system for proteins and other important materials between the nucleus and the rest of the cell?
  - a. Ribosomes
  - b. Golgi apparatus
  - c. Cytoplasm
  - d. Endoplasmic reticulum

5. Which of the following structures in a cell controls the passage of water and other materials into and out of the cell?
- a. Cell membrane
  - b. Cell wall
  - c. Lysosome
  - d. Nucleus
6. The cell wall of plants cells
- a. provides support for the plant cell
  - b. gives the cell its shape
  - c. is made up of cellulose
  - d. all of the above
7. When you use a microscope to observe cells, light passes through the specimen then
- a. is reflected off mirrors to make the image bigger
  - b. to our eye where its image is magnified
  - c. through two lenses which magnify the image of the specimen
  - d. through the diaphragm to our eye
8. Turning this knob provides the **initial** focus when using a microscope.
- a. Fine focus
  - b. Coarse focus
  - c. Eye Piece
  - d. Objective lens
9. Always **begin** observing a specimen under the microscope using the \_\_\_\_\_ objective lens.
- a. 100X
  - b. 40X
  - c. 10X
  - d. 4X

10. Which of the following statements regarding microscope use and technique is correct?
- a. When carrying the microscope, it should be supported by placing both hands under the base.
  - b. The slide may be placed anywhere on the stage, as long as there is light shining on it.
  - c. When using the coarse focus dial on the microscope, you should be looking at the side of the microscope.
  - d. If you want to see as much of the specimen as possible, you should be on high power.

**End of Section One**

## Section Two: Short Answers

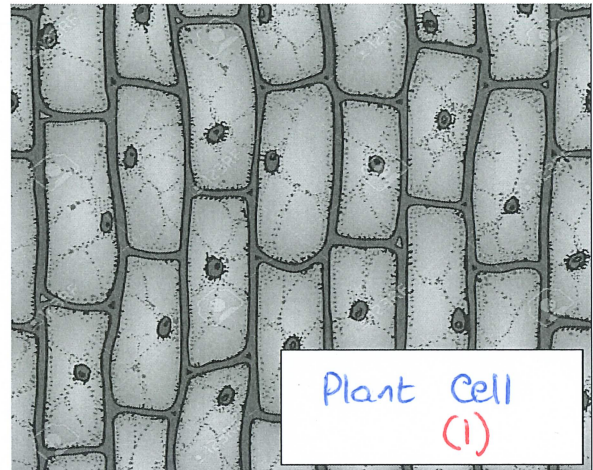
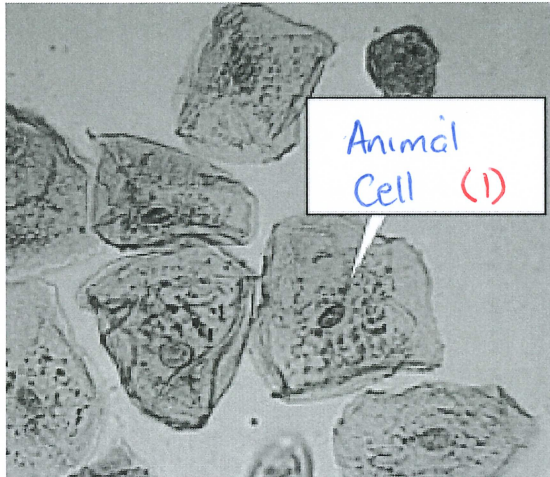
[45 Marks]

Answer ALL questions in the spaces provided below. Use a blue or black pen unless you have been asked to draw a diagram.

### Question 1

[11 Marks]

The two pictures below show illustrations of plant cells animal cells.



- a. Identify which illustration represents a typical plant cell and which is an animal cell. Write your answer in the box provided on each illustration. (2)

- b. Other than the colour of these two cells, state one reason for your answer at (a). Your reason must refer to something that is **clearly visible in the illustrations** above. (1)

(1) Plant cell has a rigid shape (due to its cell wall) while the animal cell has an irregular shape (due to the lack of a cell wall).

- c. Name **two** components that these cell types have in common. The two components must be **clearly visible in the illustrations** above. (2)

(1) Nucleus  
(1) Cytoplasm  
(1) Cell membrane. } Any 2.

Plant cells contain a chemical called chlorophyll that enables them to use the light energy from the sun to make their own food (glucose). Answer the following questions about this process in plants:

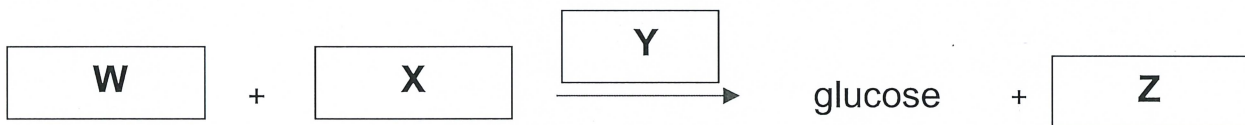
- d. What is the name of this process by which plants make food? (1)

Photosynthesis (1)

- e. In which organelle in the plant cell does this process occur? (1)

chloroplasts (1)

- f. The equation shown below represents this process. Complete the word equation by writing the names of the substances represented by each of the letters in the boxes. (4)



W: Carbon dioxide (1)

X: Water (1)

Y: Sunlight / Light energy (1)

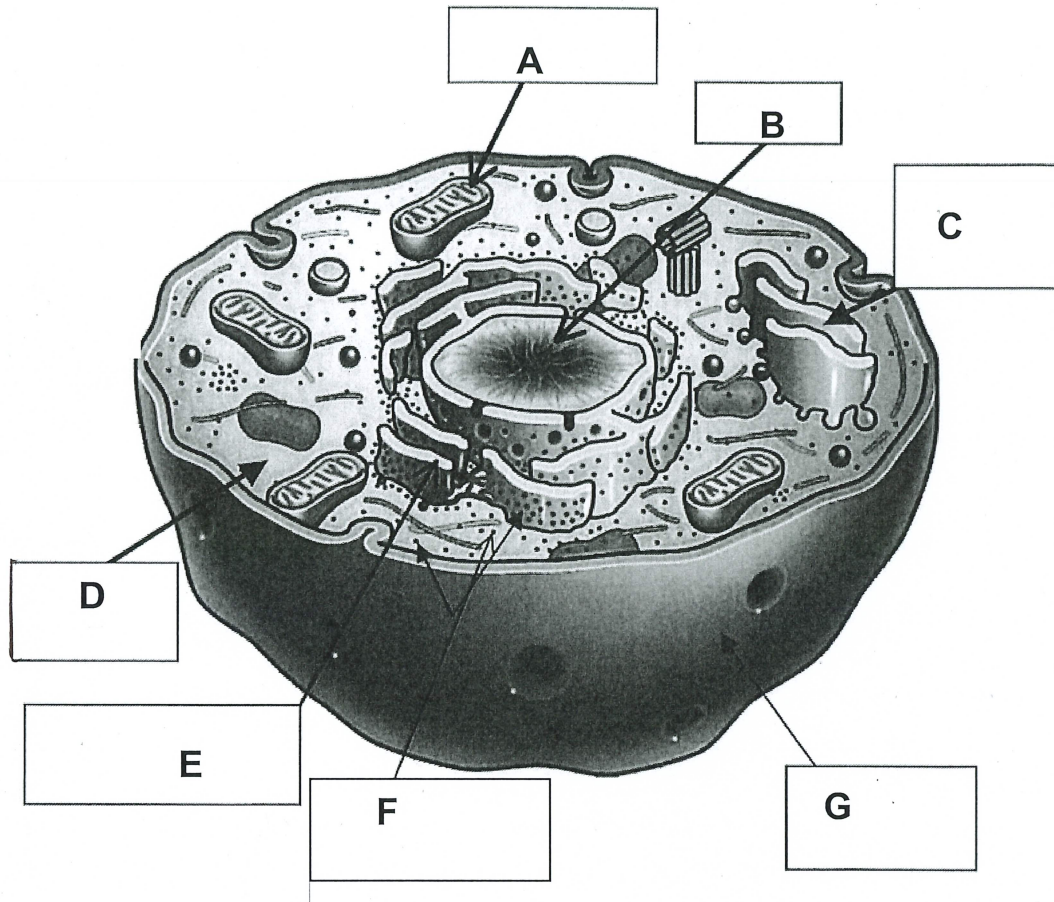
Z: Oxygen (1)

} Any order

## Question 2

[11 Marks]

Refer to the diagram below and answer the questions that follow.



a. Name the parts of the cell represented by the letters:

- |   |                                 |                             |                                  |
|---|---------------------------------|-----------------------------|----------------------------------|
| A | <u>Mitochondrion</u> (1)        | (Half if abbreviated to ER) | (7)                              |
| B | <u>Nucleus</u> (1)              | E                           | <u>Endoplasmic reticulum</u> (1) |
| C | <u>Golgi apparatus/body</u> (1) | F                           | <u>Ribosomes</u> (1)             |
| D | <u>Cytoplasm/cytosol</u> (1)    | G                           | <u>Cell membrane.</u> (1)        |

b. Which letter (A to G) on the above diagram: (2)

(i) is the control centre of the cell? B (1)

(ii) is the site where proteins are made? F (1)

c. State the function of the following parts of the cell: (2)

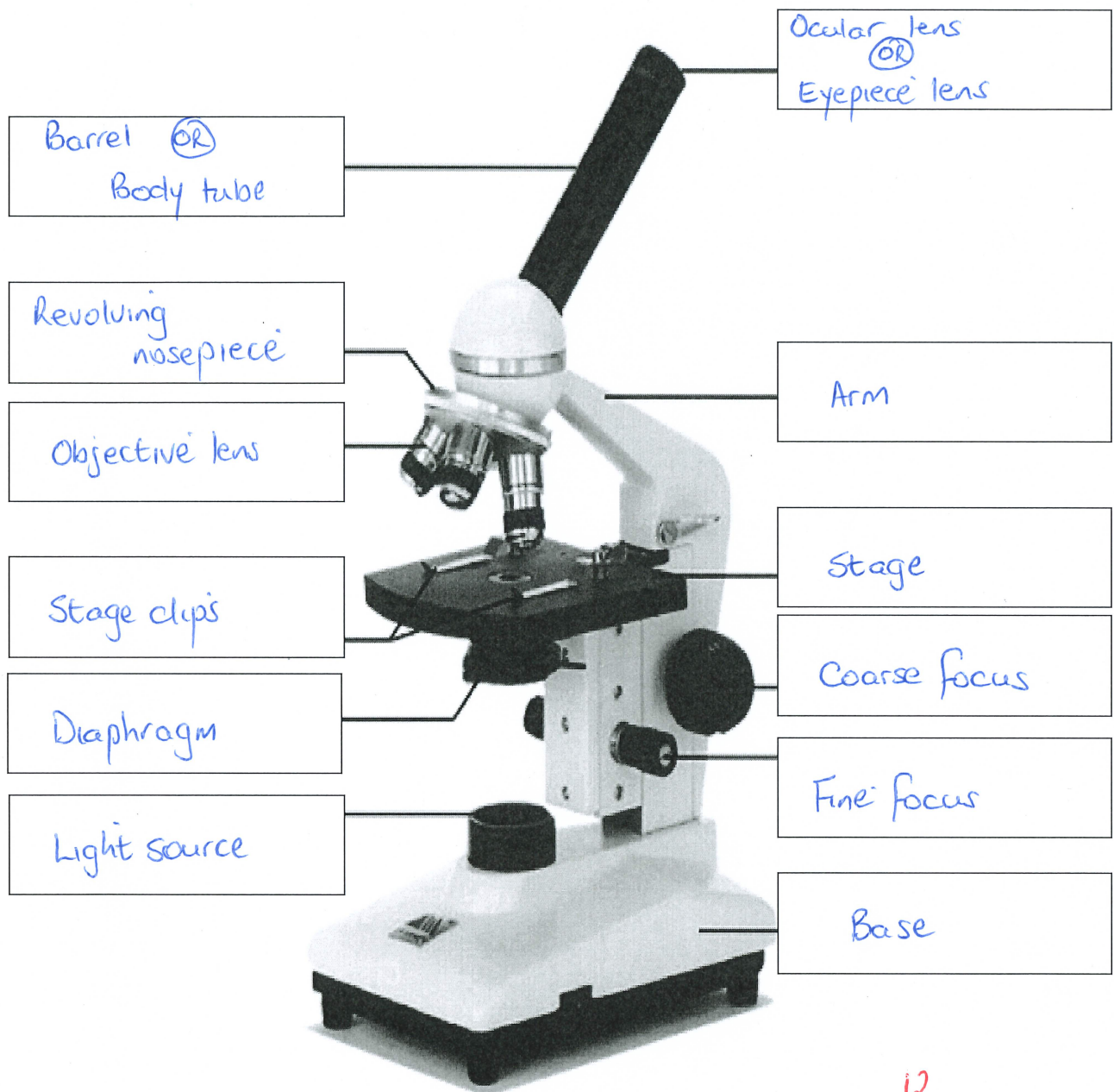
Lysosome: Contains chemicals that break down waste materials in the cell (1)

Mitochondrion: Produces energy for the cell's activities during the process of respiration (1)

### Question 3

[12 Marks]

Label the parts of the microscope represented by each of the boxes on the diagram.



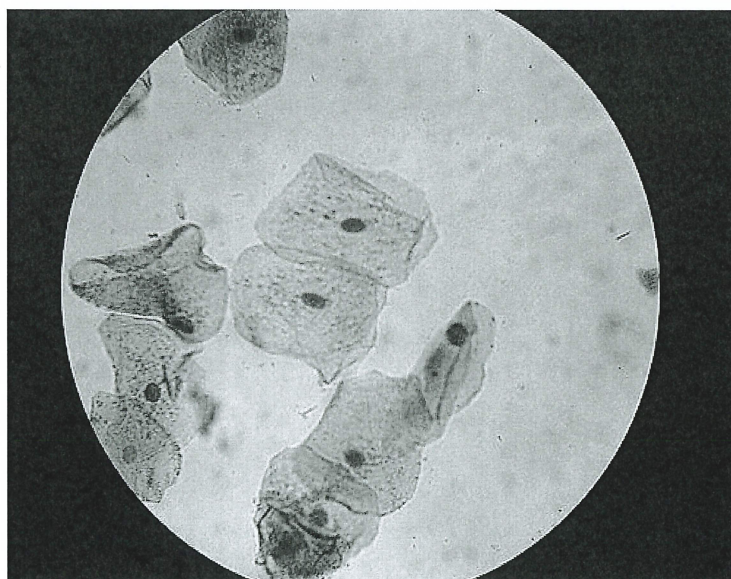
**Question 4****[11 Marks]**

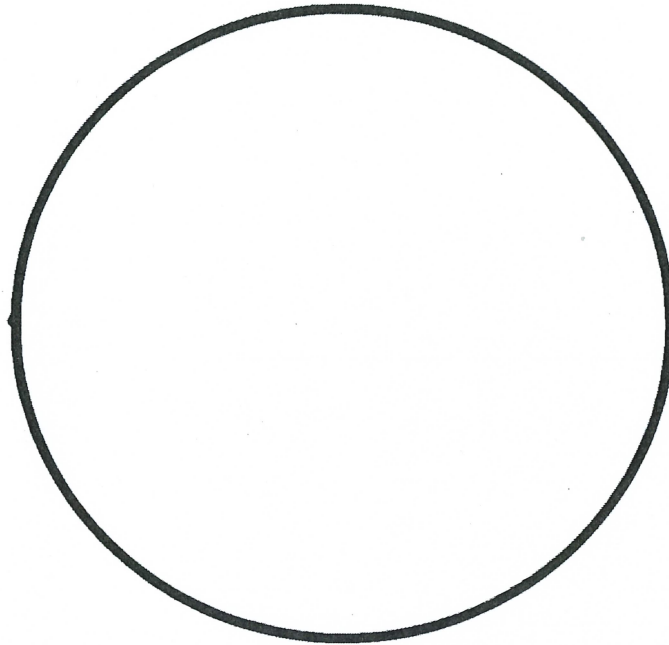
- a. Complete the table below by filling in the missing values used to determine the magnification of a light microscope. (3)

| Ocular Lens | Objective Lens | Total Magnification |
|-------------|----------------|---------------------|
| 10X (i)     | 10X            | 100X                |
| 10X         | 4X             | 40X (i)             |
| 5X          | 40X (i)        | 200X                |

- b. The picture below shows the image that was observed by a student viewing a specimen of human cheek cells, while using the 10X objective lens and a 5X ocular lens.

In the circle provided on the next page, draw a **neat, labelled diagram** to show how you would represent this image. **Remember to follow all the rules for microscope drawings.** (8)





(1) Title: Human Cheek Cells

(1) Total Mag: 50X

(1) Pencil diagram

(1) No shading / only outline of cell & its structures

(1) Diagram representative of image (i.e. no. & location of cells)

(3)\* Labels: Nucleus, cell membrane, cytoplasm (2)\*  
All out on one side with straight lines  
(no crossing over). (1)

(\* -1) if less than 3 labels

END OF TEST